

MINI PROJECT – TO IMPLEMENT ROBOT ARM CONTROL

AIM:

To write a program that simulates the control of a robotic arm with three joints (base, shoulder, elbow). The arm should move through different positions such as Pick, Place, and Home, and display the corresponding joint angles during execution.

ALGORITHM:

1. Start the program.
2. Initialize the servo motors and the serial communication.
3. Define the initial joint angles for the Home position.
4. Move all servos to the Home position and display the angles
5. Set predefined positions for Pick, Place, and Home
6. Move the arm from Home to the Pick position gradually
7. Display the angles when the Pick position is reached and simulate object pick-up
8. Move the arm from Pick to Place position in smooth steps
9. Display the angles when the Place position is reached and simulate placing the object
10. Move the arm back from Place to Home position gradually
11. Display the angles for Home position
12. Repeat the cycle continuously in loop
13. End when the system is stopped.

PROGRAM :

```
#include <Servo.h>
Servo servoBase, servoShoulder, servoElbow;
//Define positions
int pickPos[3] = {90, 120, 40};
int placePos[3] = {10, 80, 100};
int homePos[3] = {90, 60, 90};

// Current angles of the arm
int currentAngles[3] = {90, 60, 90};

void setup() {
  Serial.begin(9600);
  servoBase.attach(9);
  servoShoulder.attach(10);
  servoElbow.attach(11);

  // Move to Home at start
  moveSmooth(homePos[0], homePos[1], homePos[2]);
  Serial.println("Robotic Arm Initialized at HOME position");
}

void loop() {
  Serial.println("Moving to PICK position...");
  moveSmooth(pickPos[0], pickPos[1], pickPos[2]);
  Serial.println("Picked the object");

  Serial.println("Moving to PLACE position...");
  moveSmooth(placePos[0], placePos[1], placePos[2]);
  Serial.println("Placed the object");

  Serial.println("Returning to HOME position...");
  moveSmooth(homePos[0], homePos[1], homePos[2]);
  Serial.println("Cycle Completed\n");
  delay(2000); }

// Function to move smoothly from current to target
void moveSmooth(int targetBase, int targetShoulder, int targetElbow) {
  int steps = 30; // number of small steps
  for (int i = 1; i <= steps; i++) {
    int baseStep = currentAngles[0] + (targetBase - currentAngles[0]) * i / steps;
    int shoulderStep = currentAngles[1] + (targetShoulder - currentAngles[1]) * i / steps;
    int elbowStep = currentAngles[2] + (targetElbow - currentAngles[2]) * i / steps;
    servoBase.write(baseStep);
```

```

servoShoulder.write(shoulderStep);
servoElbow.write(elbowStep);
delay(50); // smooth motion delay
}
currentAngles[0] = targetBase;
currentAngles[1] = targetShoulder;
currentAngles[2] = targetElbow;

// Print angles after reaching target
Serial.print("Angles => Base: ");
Serial.print(currentAngles[0]);
Serial.print("°, Shoulder: ");
Serial.print(currentAngles[1]);
Serial.print("°, Elbow: ");
Serial.print(currentAngles[2]);
Serial.println("°");
}

```

OUTPUT:

Robotic Arm Initialized at HOME position
Angles => Base: 90°, Shoulder: 60°, Elbow: 90°

Moving to PICK position...
Angles => Base: 90°, Shoulder: 120°, Elbow: 40°
Picked the object

Moving to PLACE position...
Angles => Base: 10°, Shoulder: 80°, Elbow: 100°
Placed the object

Returning to HOME position...
Angles => Base: 90°, Shoulder: 60°, Elbow: 90°
Cycle Completed

RESULT:

Thus the robot arm control program was implemented and executed successfully.